

5(a)	sum of current(s) in = sum of current(s) out or (algebraic) sum of current(s) is zero	M1
	at a junction (in a circuit)	A1
5(b)(i)	(current in R_4 or R_1 =) $0.30 + 0.30$ (= 0.60 A)	B1
	(R =) $2.4 / 0.60 = 4.0 \text{ } (\Omega)$	A1
	or	
	(p.d. across R_3 or R_2 =) $2.4 / 2$ (= 1.2 V)	(B1)
	(R =) $1.2 / 0.30 = 4.0 \text{ } (\Omega)$	(A1)
5(b)(ii)	$E = 2.4 + 2.4 + 1.2$	C1
	= 6.0 V	A1
	or	
	total resistance = $10 \text{ } (\Omega)$	(C1)
	$E = 10 \times 0.60$ = 6.0 V	(A1)
5(c)	total resistance increases	B1
	current decreases (in battery) so total power decreases	B1

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Question	Answer	Marks
5(d)	resistivity = RA / L	C1
	$= 4.0 \times \pi \times (240 \times 10^{-6})^2 / 0.67$	C1
	$= 1.1 \times 10^{-6} \Omega \text{ m}$	A1

Question	Answer	Marks
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